

Team Formation + Project Scoping Studio

+ Week 04

Dr. Xuhai “Orson” Xu | Dr. Eugene Wu

AI @ VP&S Initiative



VAGELOS COLLEGE OF
PHYSICIANS AND SURGEONS

COLLABORATORY
AT COLUMBIA UNIVERSITY



Today's Outline

- 1. Revisit the Capstone Project**
- 2. Your Project Ideas - Presentation**
- 3. Guest Lecture: Dr. Ali Heydari**

A Personal Health Assistant (PHA)

A **Personal Health Assistant (PHA)** combines authorized personal health data and computational tools to help a person inspect patterns, ask questions, and prepare decisions. Its claims remain traceable to evidence and bounded by intended use.

Many current PHAs use a generative-AI agent, or a team of specialized agents, to connect users with data and tools.

This course uses **assistant** for the user-facing role and **agent** for one possible implementation

Personal Health Agent is another common term, but the broader PHA vision predates LLMs.

Our goal is to build a **Personal Health Assistant (PHA)**
whose outputs are **personalized, dynamic, inspectable, safe,**
and aligned with users.

Capstone Project

- You will form a team of **3–4**, ideally with diverse backgrounds
 - Our class has a balanced mix of health and CS students
- Each team will choose a specific target user group and question to address with a PHA system
- The system should use at least one of: (1) clinical/EHR, (2) everyday sensors, or (3) self-report source
 - We will revisit these three sources today
 - Multiple data modalities are encouraged and preferred
- Build an LLM, RAG, or agent component with a usable interface
- The system can be any reasonable format: website, mobile app, desktop app, command-line tool, or something creative...

Capstone Project

Read more about the capstone project in the course syllabus:

personal-health-agent.github.io/project-details/

Start ideation and team formation early!

- Week 4: Project Pitch & Team Formation <--- **we are here**
- Week 9: Project Proposal with Low-Fidelity Presentation
- Week 14: Project Demo and Poster; Writeup Due Dec 16

A Daily Join Keeps Three Sources Visible

Analysis grain: one person-day in local time · 2026-09-18

Clinical/EHR	Everyday sensor	Self-report
Hemoglobin: 10.8 g/dL Specimen: 09:10 Status: final Observation/obs-401	Resting HR: 75 bpm Daily estimate Wear coverage: 18.5 h Method: DERIVED	Fatigue: 7/10 at 21:05 Recall: “today” Exact item retained Note: “Slept poorly”

The date aligns records for analysis.

But always remember: It does **not** make them simultaneous or show that one caused another.

Tell Us About Your Idea

A few key questions:

- **Who is it for, and what question do they have?**
- **What data would help answer that question?**
- **What would your PHA help the target user do?**

Each idea: 2–3 minutes to present + 1 minute for Q&A.

Project Idea Pitches

2-3 minutes to present + 1-2 minute for Q&A

Break

Continue a conversation with someone whose idea interested you.

We will return for the guest lecture.

Lab: Reflect on Today's Talks

After the guest talk, write:

1. What did you learn from **classmates' project ideas** and the **guest lecture**? Use specific moments that changed or clarified your thinking.
2. How did you use AI for this reflection? If you did not use AI, say so.

Complete the TODOs in the Week 4 notebook.

Guest Lecture

Dr. Ali Heydari

Senior Research Scientist, Google Research

From Passive Sensing to Actionable Futures: Towards Autonomous Personal Health Agents Powered by Physiological World Models

Friday, October 2 · Online · Final hour of class

[Talk details on the course website](#)

Thanks for the Attention
Q&A